Giwon Baek

Seoul, Republic of Korea
Mobile: +82 010-5268-9514
Email: 01gwbaek@naver.com
GitHub: github.com/BBaekGiwon
Portfolio: (Link)



Personal Profile

Robotics researcher/engineer with hands-on experience in vision-language-action models, dexterous manipulation, teleoperation, and robot learning. Experienced in developing end-to-end robotics pipelines from data collection and model training to simulation and real-robot deployment using Python, PyTorch, ROS 2, and NVIDIA Isaac Sim/Gym.

Education

Mar 2020 – Feb 2027: B.S. in Computer Science and Engineering

Sogang University, Seoul

Relevant Coursework: Computer Vision, Basic Machine Learning, Deep Learning, Introduction to Artificial Intelligence, Operating Systems.

Experience

Oct 2026 – Ongoing: Robotics Deployment Engineer Intern

RLWRLD RD department, Seoul

- coming soon.

Aug 2026 – Oct 2026: Robotics Research Intern

Korea University RI Lab, Seoul

- Participating in RLWRLD projects on vision-language-action (VLA) learning for humanoid manipulation.
- Developing data collection and training pipelines for bimanual manipulation using teleoperation demonstrations.
- Exploring VLM-based automated labeling of subtask instructions for VLA training data.
- Preprocessing VLA trajectory data by removing low-motion segments.

Dec 2025 – Jun 2026: Robotics Research Intern

Korea Institute of Science and Technology (KIST) HARI Lab, Seoul

- Developed affordance-aware dexterous grasp generation pipelines from 3D object point clouds using vision-language models and part-aware segmentation.
- Applied a cross-embodiment model to transfer dexterous manipulation policies to a KISTAR hand, a custom robotic hand developed at KIST.
- Collected humanoid teleoperation data for dexterous manipulation tasks using VR-based teleoperation systems.
- Constructed dexterous grasp pose datasets for the KISTAR hand using Optimization open-source models, and Isaac Gym-based validation.

Sept 2025 – Dec 2025: Like Lion AI CV Bootcamp

Like Lion, Seoul

- Completed intensive training in computer vision and deep learning.
- Improved RT-DETR-based object detection performance for autonomous driving scenarios using a Vision Mamba encoder.
- Enhanced medical image captioning performance by integrating segmentation masks with RGB breast tissue images.
- Developed a real-time deepfake detection video call application. Also received first place in the KEIT AI Life Solution Challenge.

Mar 2025 – Sept 2025: Doosan Robotics ROKEY Bootcamp

Doosan Robotics, Seoul

- Completed intensive training in robotics software, ROS 2, computer vision, and autonomous navigation.
- Developed a lane-following autonomous driving system for TurtleBot3 using a downward-facing camera,

and experienced digital twin deployment from Gazebo simulation to real robots.
- Built an autonomous mobile robot pipeline using TurtleBot4, RGB-D cameras, LiDAR, YOLO-based object detection, SLAM, and Nav2 navigation.
- Developed a voice-guided kitchen assistant robot using YOLO segmentation, PCA-based grasp orientation estimation, and MediaPipe-based hand gesture interaction.

Jun 2023 – Jul 2023 & Jan 2024 – Feb 2024: Software Engineer Intern
Learning Lab Co., Seoul
- Participated in government-funded smart manufacturing and Data voucher projects.
- Developed software pipelines for real-time manufacturing KPI collection and deployed sensor-based production monitoring systems in industrial environments.
- Collected and annotated segmentation datasets for sign language recognition and automotive defect detection tasks.
- Provided on-site deployment, maintenance, and technical support for industrial clients.

Projects

RLWRLD Conveyor Demo | 2026
Korea Universisty RI Lab
- Implemented a humanoid parcel-handling task involving retrieval, flipping, and sorting.
- Collected bimanual teleoperation demonstrations and used them to fine-tune the RLDX-1 vision-language-action (VLA) model.
- Automated subtask instruction labeling for VLA training data using vision-language models.

Development of Tactile Intelligence in Robotic Hands Based on Tactile Data Learning for Manipulating Irregular Multiple Types of Objects | 2026
Korea Institute of Science and Technology (KIST) HARI Lab
- Developed an affordance-aware dexterous grasp generation pipeline from segmented 3D point clouds.
- Made a modular grasping framework with integrating Large-Language models, part-aware segmentation, and grasp pose generation.
- Generated training data using NVIDIA Isaac Gym and deployed the system on a real robot using ROS 2.

Graspium | 2026
Independent Research Seminar (Website)
- Participated in weekly paper seminars on dexterous manipulation.
- Discussed topics including dexterous grasping, multimodal VLA, reinforcement learning, and imitation learning.
- Presented papers including AffordGrasp, D(R,O) and UniMorphGrasp.

Hecto AI Challenge | 2026
Independent Team Project
- Developed a deepfake detection model based on DINOv3.
- Constructed a large-scale dataset containing 0.6 million images for model training and preprocessing.
- Awarded second place.

Korea Planning & Evaluation Institute of Industrial Technology (KEIT) AI Life Solution Challenge | 2025
Like Lion AI CV Bootcamp
- Developed a Flutter-based video call application with CNN-based real-time deepfake detection.
- Performed server-side SigLIP analysis on suspicious frames and generated explainable reports using Grad-CAM visualizations.
- Awarded first place (Chairman's Award of the KEIT Evaluation Committee).

LG Aimers 5th | 2024
LG Aimers Program
- Completed training in machine learning, deep learning and industrial applications of artificial intelligence.
- Developed an AI model to detect defective products using sensor data from the LG Display manufacturing process.
- Ranked in the top 6% among more than 750 participating teams.

Smart Manufacturing Innovation Support Program | 2024
Learning Lab Co., Ltd.
- Designed and built a counting sensor control box to measure production volume in manufacturing environments.
- Deployed sensor systems in factories and transmitted production data to office servers for KPI monitoring.

- Developed an automated pipeline to upload KPI data to government platforms through API integration.
- Contributed to the team's first-place ranking in the interim evaluation.

Data Voucher Project | 2023
Learning Lab Co., Ltd.

- Participated in government-funded AI dataset construction projects.
- Collected and preprocessed segmentation datasets for sign language tasks.
- Constructed defect segmentation datasets for automotive component inspection tasks.

Awards

- First Place, KEIT AI Life Solution Challenge (Chairman's Award), 2025
- Second Place, Hecto AI Challenge, 2026

Skills

Programming: Python, C/C++
Frameworks: PyTorch, ROS 2, NVIDIA Isaac Sim, NVIDIA Isaac Gym
Tools: Git, Docker, Linux, Notion

Languages

- English: Fluent, OPIC AL
- Korean: Native
- Portuguese: Intermediate